LECTURE-19

Page 1: The Cell Cycle and its Phases

This page introduces the cell cycle as the life cycle of a cell, focusing on the decision to divide and the commitment points within the phases. Cells that decide to remain cycling through the cell cycle are capable of **proliferation**[4].

Interphase (Duplication of Cellular Content) [5]

Interphase is the preparation stage for cell division and consists of three sub-phases:

1. **G1 Phase (G1)**: The cell asks the question: "Should I divide?"[6].
    - **Decision Window:** This is a crucial point where the cell determines its path[7].
    - **S phase entry commitment point:** If the cell receives signals to divide, it passes this point and commits to DNA replication[8].
2. **S Phase (S)**: The cell is actively **doubling its DNA** for daughter cells[9].
    - **Activity: DNA replication** (Replicating DNA)[10101010].
    - **Activity: Centriole duplication**[11].
3. **G2 Phase (G2)**: The cell asks: "In addition to DNA, do I have everything else to make 2 daughter cells?"[12].
    - **Mitotic entry commitment point:** Passing this point commits the cell to mitosis[13].

M Phase (Mitosis - Segregation of Cellular Content) [141414]

This is the process of cell division. The cell is actively "dividing into 2"[15].

- **Prophase**[16]: Preparation for chromosome separation begins.
- **Metaphase** [17]: **Duplicated chromosomes** [18]line up, and the **mitotic spindle** [19] is fully formed.

- **Decision Window:** The cell checks if the DNA will be allocated equally to both daughters[20].
- **Commitment Point (SAC): Mitotic exit commitment point** [21] is reached upon successful check, which is the **Spindle Assembly Checkpoint (SAC)**[22].

- **Anaphase** [23]: **Chromatid separation** [24] occurs, pulling the duplicated chromosomes apart.
- **Cytokinesis** [25]: **Cellular separation** (cytoplasm division) completes, yielding two daughter cells[26].

$\text{G}0$ Phase (Quiescence/Exit)

Cells can **exit the cell cycle** [27] and enter the **$\text{G}0$** phase [28], meaning they are "not going to divide any more"[29]. A cell in $\text{G}0$ can:

- **Differentiate**[30].
- **Remain quiescent**[31].
- **Die**[32].

Page 2: Cell Cycle Checkpoints and Outcomes

This page emphasizes that progression is controlled by mandatory **checkpoints** [33] and introduces the consequences of failure.

The Checkpoint System

- **Function:** Progression through the cell cycle requires successfully "passing" several checkpoints[34].
- **Failure Outcome 1 (Stalling):** Failure to clear a checkpoint results in **stalling** at that checkpoint till repair can finish to the satisfaction of passing the checkpoint[35].
- **Failure Outcome 2 (Cell Death): Failure to repair** and pass the checkpoint results in **cell death** (Apoptosis)[36].

Checkpoints and Regulatory Complexes (Table)

The table links key cell cycle events to the regulatory proteins and the consequences of failure[37].

Cell Cycle Phase	Checkpoint	Key Regulators	Outcome of Failure (Increasing levels of DNA damage)	
Interphase (Exit $\to$ Entry)	**DNA damage checkpoint**	**Cyclin D, $\text{CDK}4/6$** (Block cell cycle entry)		**Quiescence**[38], **Senescence**[39]
S phase	**Replication stress checkpoint**	**Cyclin E/A, $\text{CDK}2$** (Block replication initiation)		**Senescence**[40]
M phase (Entry)	**DNA damage checkpoint**	**Cyclin A/B, $\text{CDK}1/2$** (Block mitotic entry)		**Apoptosis**[41]
M phase (Metaphase)	**Spindle assembly checkpoint**	**APC/C** (Block mitotic exit)		**Apoptosis**[42]

Page 3 & 4: CDK/Cyclin Mechanism and the Cell Cycle Algorithm

These pages provide a detailed molecular view of cell cycle progression, viewing the cycle as an **algorithm**[43].

The Checkpoint Machinery

- **Checkpoints** are composed of **protein complexes** of **Cyclin Dependant Kinases (CDKs)** and **Cyclins**[44].
- **Cyclins:** Their protein levels **rises and falls** with phases of the cell cycle[45].
- **CDKs:** These are **enzymes** present at all phases but are **not active unless partnered with a Cyclin**[46].
- **Activation & Progression:**
    1. Upon partnering with a Cyclin, the CDK-Cyclin pair **phosphorylates** other proteins[47].
    2. The phosphorylated proteins get **degraded**[48].
    3. This degradation allows the cell to **progress to the next phase** and checkpoint[49].

Molecular Steps (Algorithm) of the Cell Cycle

The figure illustrates the main molecular steps and the phases they control:

- **$\text{G}0$ (Prereplicative State):** The cell is prevented from exiting the cell cycle by the action of **Cyclin D** with **$\text{CDK}4/6$**[50505050]. If **favourable conditions** are absent, the cell cycle exits[51].
- **$\text{G}1$/Decision Window:**
    - **RB** [52] acts as an inhibitor.
    - **Cyclin E** [53] and **$\text{CDK}2$** [54] phosphorylate RB to **remove inhibitors**[55], allowing the cell to enter the cell cycle.
    - This leads to **$\text{E}2\text{F}$-dependent transcription** [56565656] and the cell expands material[57].

- **S Phase:**
  - **Cyclin A** [58] with **$\text{CDK}2$** [59] facilitates **replication initiation**[60606060].
- **$\text{G}2$ (Postreplicative State):**
  - **Cyclin A** [61] partners with **$\text{CDK}1$** [62] to prepare for mitosis[63].
- **M Phase Entry:**
  - The complex of **Cyclin B** [64] and **$\text{CDK}1$** [65] promotes **mitotic entry**[66]. This complex ensures the cell assembles the spindle to divide the doubled DNA into two cells[67].
- **M Phase Exit:**
  - The **APC/C** (Anaphase Promoting Complex) [68] triggers the **mitotic exit** [69], allowing the cell to divide the DNA and cytoplasm into two cells[70].

The Algorithm Analogy

- The cell cycle is an **algorithm** with specific and logical steps to achieve the outcome of whether to divide or not[71].
- **Good Outcome:** The algorithm **checks for bugs** and **stalls** the algorithm until the bug gets fixed[72]. **Failure to fix the bug** results in **failure of the algorithm and cell death** (a good outcome)[73].
- **Bad Outcome: Failure to detect the bug** and/or **failure to fix the bug** results in the algorithm leading to unexpected outcomes, such as **cancer** (a bad outcome)[74].

Page 5 & 6: Cell Cycle Bugs and Cancer Treatment Strategies

These pages discuss how "unfixable bugs" in the cell cycle algorithm relate to cancer and how this knowledge is exploited for therapy.

Proteins Mutated in Cancer (Page 5 Table)

Unfixable bugs (mutations) in the cell cycle algorithm tend to occur more frequently in some parts of the code than in others[75].

Cell Cycle Control	**Proteins Frequently Mutated in Cancer**	**Proteins Rarely Mutated in Cancer**		
DNA damage/G1		ATM, $\text{CHK}2$, $\text{p}53$ [76]		$\text{G}1/\text{S}$ CDKs, Pocket proteins, $\text{E}2\text{F}$ [77]
Replication Stress/S		ATR, $\text{CHK}1$ [78]		M phase CDKs [79]
Growth/Mitogens		RAS, RAF, MYC [80]		APC/C [81]
G1/S Checkpoint		$\text{CDKN}2\text{A}$ ($\text{p}16$ and ARF) [82]		

Therapeutic Strategies Exploiting the "Bug" (Page 6)

Unfixable bugs in the cell cycle algorithm are also **opportunities** for treatment[83]. The goal is to either fix the bug, alter the bug, or take advantage of the bug to force the cell into catastrophic damage and death[8484].

Strategy	Action	Target/Mechanism	Example Drugs	Result		
Force Cell Cycle Exit	Fix the bug or alter it to end without damage [85]	Prevent cell cycle progression [86]		$\text{CDK}7\text{i}$, $\text{CDK}4/6\text{i}$ [87]		Proliferation $\downarrow$ [88888888]
Induce Cell Cycle Progression	Take advantage of the bug to force accelerated damage [89]	Override $\text{G}1$ or $\text{G}2$ arrest		$\text{WEE}1\text{i}$, $\text{p}21\text{i}$ [90]	Genome instability $\to$ Cell death [91919191]	
Induce DNA Damage	Take advantage of the bug [92]	Directly damage DNA [93]		Radiotherapy, Chemotherapy [94]	Catastrophic DNA damage levels $\to$ Cell death [95]	
Induce Replication Stress	Take advantage of the bug [96]	Cause stress during S phase [9797]		Chemotherapy [98]	DNA replication stress $\to$ Cell death [99999999]	
Impair Replication-Stress Tolerance	Alter the bug to cause damage [100]	Block the RSC response [101]		ATRi, $\text{CHK}1\text{i}$ [102]		DNA damage response impairment

						$\to$ Cell death [103103103103]
Induce Mitotic Defects	Take advantage of the bug [104]	Disrupt the mitotic spindle [105105]		Taxanes, Vinca alkaloids [106]	Mitotic defects $\to$ Cell death [1071071071 07]	
Impair SAC	Alter the bug to cause damage [108]	Block the SAC response [109]		$\text{MPS}1\text{i}$ [110]	Genome instability $\to$ Cell death [111111111111]	